Cite as:

Rane, Nitin and Choudhary, Saurabh and Rane, Jayesh (2023) Explainable Artificial Intelligence (XAI) approaches for transparency and accountability in financial decision-making. http://dx.doi.org/10.2139/ssrn.4640316

Explainable Artificial Intelligence (XAI) approaches for transparency and accountability in financial decision-making

[*1] Nitin Liladhar Rane [2] Saurabh P. Choudhary [3] Jayesh Rane

[*1,2,3] University of Mumbai, Mumbai, India
[*1] Email: nitinrane33@gmail.com

Abstract:

Recently, there has been a growing trend in incorporating Artificial Intelligence (AI) into financial decision-making, prompting concerns about the transparency and accountability of these intricate systems. This study investigates the impact of Explainable Artificial Intelligence (XAI) approaches in alleviating these concerns and improving transparency in financial decision-making processes. The paper commences by outlining the current landscape of AI applications in finance, underscoring the complex and opaque nature of advanced machine learning models. The lack of interpretability in these models presents a significant challenge, as stakeholders, regulators, and end-users often struggle to comprehend the reasoning behind AI-driven financial decisions. This opacity raises questions regarding accountability and trust, particularly in critical financial scenarios. The primary focus of the research centers on the analysis and implementation of XAI techniques to introduce transparency into financial AI systems. Various XAI methods, including rule-based systems, model-agnostic approaches, and interpretable machine learning models, are scrutinized for their effectiveness in producing understandable explanations for AI-driven financial decisions. The paper explores how these approaches can be tailored to meet the distinct requirements of the financial domain, where interpretability is essential for regulatory compliance and stakeholder confidence. Moreover, the research delves into the potential impact of XAI on accountability mechanisms within financial institutions. By offering interpretable explanations for model outputs, XAI not only enhances transparency but also empowers financial professionals to identify and rectify biases, errors, or unethical behaviour in AI algorithms. By promoting transparency and accountability, XAI not only addresses ethical concerns but also facilitates the responsible and trustworthy deployment of AI in the financial sector. This, in turn, contributes to the advancement of fair, reliable, and secure financial systems.

Keywords: Explainable Artificial Intelligence, Finance, Explainable AI, Risk assessment, Forecasting, Decision-making.

Introduction

In recent years, a significant transformation has occurred in the financial landscape as Artificial Intelligence (AI) and machine learning (ML) have been integrated into decision-making processes [1-4]. Financial institutions increasingly rely on advanced algorithms to analyze extensive data sets and inform critical decisions [5-8]. Consequently, the demand for transparency and accountability in these automated systems has become crucial [9-12]. Explainable Artificial Intelligence (XAI) has emerged as a pivotal area of research, seeking to bridge the gap between the complexity of advanced AI models and the comprehensibility required for effective decision-making in finance [13-18]. Financial decisions, from credit scoring to investment strategies, are now frequently entrusted to AI systems capable of processing data at unprecedented speeds. However, the inherent opacity of many state-of-the-art AI models poses challenges for stakeholders seeking to understand, interpret, and trust the outcomes of these automated decisions. The repercussions of biased or inaccurate AI-driven financial decisions can be severe, impacting both individual consumers and the stability of financial markets. This paper explores the realm of XAI and its crucial role in enhancing transparency and accountability in financial decision-making processes.

The adoption of AI in financial institutions promises improved efficiency, reduced costs, and enhanced decision-making capabilities [3,10]. However, the intricate nature of advanced AI models, such as deep neural networks, often renders them inscrutable, resulting in the "black box" problem. Stakeholders, including regulators, investors, and consumers, grapple with understanding the rationale behind AI-generated decisions. XAI has emerged as a response to the need for transparency and accountability in AI systems [14,17]. It focuses on developing algorithms and methodologies that provide insights into the decision-making processes of complex models, making them more understandable to human users. In financial decision-making, XAI aims to enable stakeholders,

Electronic copy available at: https://ssrn.com/abstract=4640316

including regulators, auditors, and end-users, to comprehend how AI models arrive at specific conclusions and predictions. Various XAI approaches, from rule-based systems to model-agnostic methods like LIME and SHAP, have been proposed and developed, enhancing the applicability of XAI across a wide range of financial applications [19-27]. Transparency is a cornerstone of trust in financial systems. As AI algorithms become increasingly intertwined with financial decision-making, XAI provides the means to uncover the decision-making process, offering transparency and fostering trust in the outcomes of AI-driven financial decisions. Moreover, transparency in financial decision-making is not only a regulatory imperative but also a strategic advantage for financial institutions [28-35]. Clear and interpretable AI models empowers institutions to demonstrate compliance with regulatory requirements, mitigate legal risks, and enhance their reputation. Figure 1 shows the co-occurrence analysis of the keywords in literature.

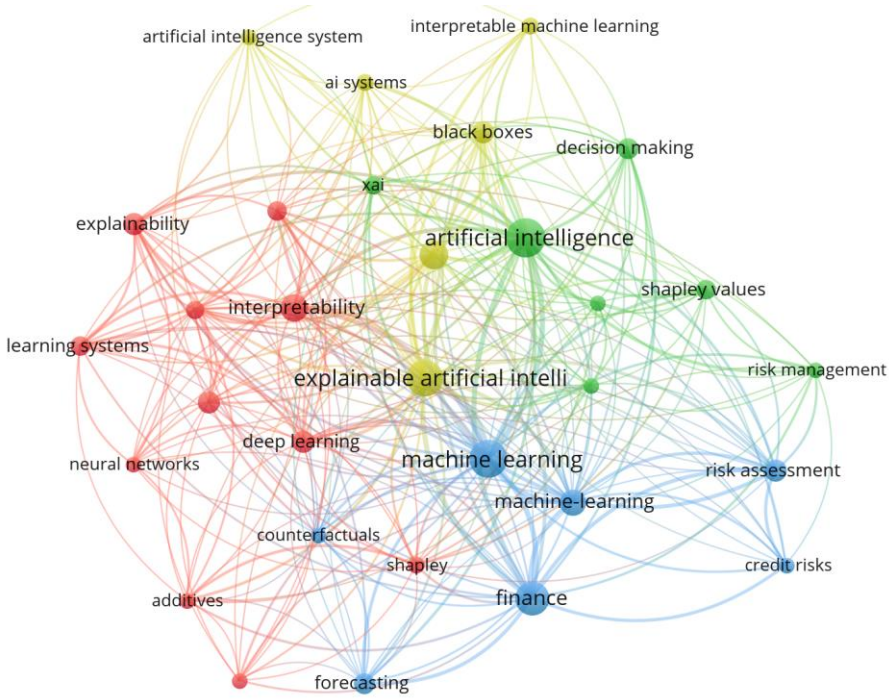

Figure 1 Co-occurrence analysis of the keywords in literature

Deploying AI in financial decision-making poses the challenge of potential bias in algorithmic outcomes [36-40]. XAI plays a crucial role in identifying and mitigating bias by providing interpretable insights into the factors influencing model decisions. Techniques like fairness-aware machine learning and algorithmic auditing enable financial institutions to systematically evaluate the impact of their models on different demographic groups, contributing to a more inclusive and socially responsible financial ecosystem. Recognizing the growing influence of AI in financial decision-making, regulatory bodies worldwide are adapting to ensure ethical standards and legal frameworks are followed. XAI has emerged as a key enabler of regulatory compliance, providing a means to interpret, validate, and audit AI models to ensure adherence to regulatory requirements [15,17]. This research paper explores various XAI approaches and their application in the financial sector, contributing to the ongoing dialogue on responsible AI deployment in the financial industry. The adoption of XAI approaches stands as a crucial step towards ensuring ethical, transparent, and accountable financial decision-making processes as financial institutions navigate the evolving landscape of AI.

Methodology

The literature review delves into a systematic and exhaustive exploration of academic databases, journals, and conference proceedings. Key search terms, such as "Explainable Artificial Intelligence," "XAI," "transparency," "accountability," and "financial decision-making," guide the search process. Inclusion and exclusion criteria are established to ensure the relevance and quality of the selected literature, with a focus on recent publications to

Electronic copy available at: https://ssrn.com/abstract=4640316

capture the latest advancements in XAI within the financial domain. During the review, emphasis is placed on identifying and categorizing relevant studies, theoretical frameworks, and practical implementations of XAI in financial contexts. Special attention is given to methodologies and approaches that enhance transparency and accountability, addressing challenges linked to the opacity of complex AI models in financial decision-making. The bibliometric analysis, on the other hand, quantitatively assesses the scholarly output in the field of XAI in financial decision-making. Utilizing the Scopus and Web of Science databases, the analysis includes extraction and examination of bibliometric data such as citation counts, publication trends, authorship patterns, and keyword co-occurrences, using Vosviewer software. The objective of the analysis is to pinpoint influential publications, prolific authors, and seminal works within the domain. Network analysis reveals connections and collaborations among researchers and institutions, while clustering algorithms identify thematic clusters and emerging topics in the field of XAI in financial decision-making. The findings from both the literature review and bibliometric analysis are then synthesized to present a comprehensive overview of the current state of XAI approaches for transparency and accountability in financial decision-making. This synthesized data is interpreted in the context of existing theories, practical applications, and the evolving landscape of AI in finance. Figure 2 shows the co-authorship analysis.

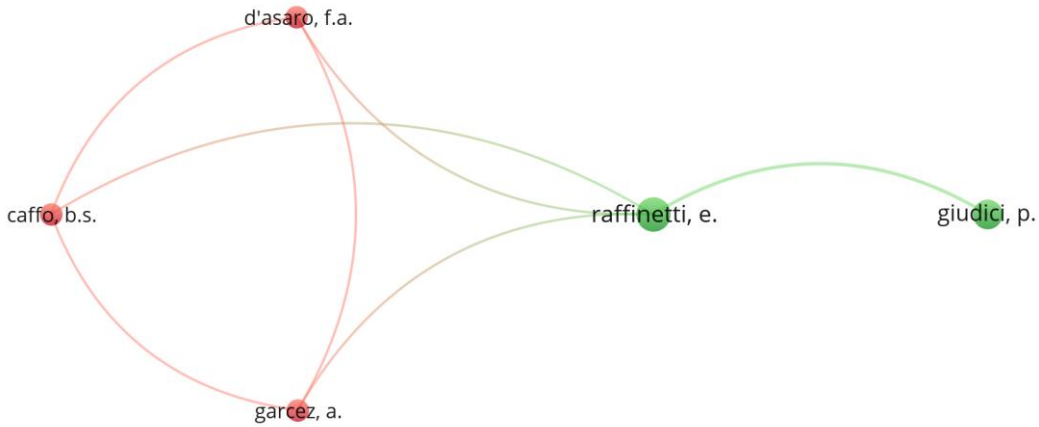

Figure 2 Co-authorship analysis

Results and discussion

Model-Specific Explainable AI approaches for financial decision-making

The focus of model-specific explainability lies in offering insights into the internal mechanisms of a specific AI model, elucidating its decision-making process [14-16]. Complex algorithms, such as deep neural networks and ensemble methods, often achieve high accuracy but lack interpretability. The opaqueness of these models poses challenges in understanding and justifying their decisions, particularly in situations where regulatory requirements demand transparency. This section explores diverse model-specific XAI approaches customized for financial decision-making. Table 1 shows the Explainable Artificial Intelligence (XAI) approaches in financial decision-making.

Decision Trees and Rule-Based Models:

Decision trees inherently offer interpretability by recursively partitioning the input space based on feature values. In financial decision-making, decision trees find applications in modeling credit scoring, fraud detection, and risk assessment [41-45]. Each node in the tree represents a decision based on a specific feature, providing a clear and intuitive decision path. Rule-based models, akin to decision trees, generate easily understandable decision rules typically in the form of "if-then" statements. This format makes them interpretable to both experts and non-experts, a valuable trait in financial domains where regulatory bodies often require institutions to justify their decisions.

Linear Models:

Electronic copy available at: https://ssrn.com/abstract=4640316

Linear models, such as linear regression and logistic regression, have been widely employed in finance due to their simplicity and interpretability. Assuming a linear relationship between input features and output, such models facilitate a straightforward analysis of the impact of each feature on the decision [46-49]. Coefficients assigned to each feature indicate the direction and magnitude of influence, aiding in risk assessment and portfolio optimization. Despite their simplicity, linear models may fall short in capturing complex non-linear relationships present in data [47,48,50-54]. In such cases, feature engineering or more complex models may be necessary.

Local Interpretable Model-agnostic Explanations (LIME):

LIME, as a model-agnostic approach, explains the predictions of any machine learning model by approximating its decision boundary locally. In financial decision-making, LIME can be applied to black-box models to generate understandable explanations for individual instances [20,23]. LIME achieves this by perturbing the input data and observing the corresponding changes in the model's predictions. It then fits a locally interpretable model, such as a linear model, to explain the model's behavior around the specific instance. This approach provides valuable insights into why a model made a particular decision, enabling stakeholders to comprehend and trust the model's output.

SHapley Additive exPlanations (SHAP):

SHAP values, rooted in cooperative game theory, allocate contributions of each feature to the model's output. In financial decision-making, SHAP values help understand the impact of individual features on a specific prediction or decision [25,27]. By assigning a Shapley value to each feature, one can quantify its contribution to the model's output, enhancing interpretability. SHAP values offer a comprehensive view of feature importance, considering interactions between features. This is crucial in finance, where intricate relationships between economic indicators can significantly influence outcomes. Understanding these interactions aids in risk assessment and scenario analysis.

The important equation used in explainable artificial intelligence (XAI) is the SHapley Additive exPlanations (SHAP) values formula. The SHAP values provide a way to fairly allocate contributions among features in a prediction model. The formula is given by:

$$\phi_i(f) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!\,(|N| - |S| - 1)!}{|N|!} [f(S \cup \{i\}) - f(S)]$$

Where,

$\phi_i(f)$ represents the SHAP value for feature i in the model f.

N is the set of all features

S is a subset of features excluding i

$f(S)$ is the model's prediction when considering only the features in S

$f(S \cup \{i\})$ is the model's prediction when adding feature i to S.

Tree-based Ensembles and Feature Importance:

Ensemble models like Random Forests and Gradient Boosting are potent tools in finance owing to their high predictive accuracy [55-60]. While the ensemble itself may be complex, analyzing the individual decision trees within it allows deriving feature importance. Feature importance in ensemble models is often measured by how much a particular feature contributes to the reduction of impurity (e.g., Gini impurity) in the decision trees [61-68]. Features with higher importance values play a more significant role in the model's decision-making process. Communicating these importance scores to stakeholders helps them understand which factors are critical in driving model predictions. Figure 3 shows the Explainable Artificial Intelligence (XAI) approaches for transparency and accountability in financial decision-making.

Electronic copy available at: https://ssrn.com/abstract=4640316

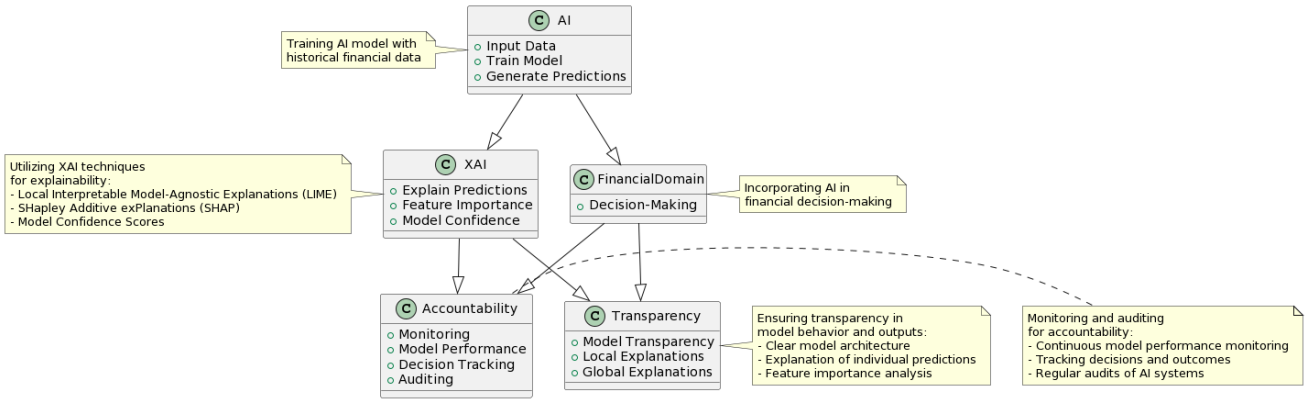

Figure 3 Explainable Artificial Intelligence (XAI) approaches for transparency and accountability in financial decision-making

Model-Agnostic Explainable AI approaches for financial decision-making

Model-Agnostic Explainable AI (XAI) methodologies play a pivotal role in enhancing transparency and interpretability within financial decision-making processes [69-73]. As machine learning like models grow in complexity, the necessity to comprehend the reasoning behind their predictions becomes increasingly vital, particularly in high-stakes domains [3,5,74-78]. Model-Agnostic XAI encompasses techniques and methodologies applicable to interpreting predictions from any machine learning model, irrespective of its underlying architecture. This approach proves especially pertinent in scenarios where black-box models, such as deep neural networks or ensemble methods, are deployed. Despite their predictive accuracy, these models often lack transparency, posing challenges for stakeholders in grasping the factors influencing their decisions. The objective of Model-Agnostic XAI is to furnish meaningful insights into model predictions, enabling users to trust, validate, and act upon the model's outputs. This is particularly crucial in the financial industry, where decisions have far-reaching impacts on investments, risk management, and overall business strategies.

Within the financial sector, decision-making heavily relies on accurate predictions and risk assessments [14,17]. However, conventional machine learning models, including intricate algorithms like deep neural networks, are frequently treated as "black boxes" due to their complex internal structures. This lack of interpretability raises concerns regarding accountability, ethics, and the potential for biased outcomes. Model-Agnostic XAI addresses these challenges by offering transparency into the decision-making process. This transparency is imperative for financial institutions to adhere to regulations, mitigate risks, and foster trust with clients and stakeholders. By understanding the factors influencing predictions, users can make more informed decisions, validate model outputs, and identify potential biases.

LIME (Local Interpretable Model-agnostic Explanations):

LIME is a popular approach that generates locally faithful explanations for model predictions [20,23]. It achieves this by perturbing the input features of a given instance and observing changes in the model's output. By fitting a locally interpretable model to the perturbed instances, LIME provides insights into how the model behaves around a specific data point. In finance, LIME can elucidate individual predictions, aiding in understanding the model's decision-making process for specific transactions or investment recommendations.

SHAP (SHapley Additive exPlanations):

SHAP values, grounded in cooperative game theory, attribute each feature's contribution to the difference between the model's prediction and the average prediction [24-27]. These values offer a fair way to distribute the model's output among its features. In financial decision-making, SHAP values can be employed to attribute the impact of different factors to a specific prediction, helping stakeholders understand the relative importance of various features in investment or risk assessment.

Electronic copy available at: https://ssrn.com/abstract=4640316

Table 1 Explainable Artificial Intelligence (XAI) approaches in financial decision-making

Sr No	XAI Approach	Description	Advantages	Challenges	Applications
1	Rule-Based Systems	Decision-making governed by predefined rules	Offers clear transparency	Limited handling of complex scenarios, potential accuracy issues	Credit Scoring, Regulatory Compliance
2	Decision Trees	Tree-like models illustrating decision pathways	Easy to comprehend	Prone to overfitting	Fraud Detection, Loan Approval
3	Linear Models	Decisions based on linear relationships	Simplicity in interpretation	Limited to linear relationships	Risk Assessment, Portfolio Management
4	LIME (Local Interpretable Model-agnostic Explanations)	Generates interpretable models for specific instances	Model-agnostic, applicable to black-box models	Global model behavior may not be fully captured	Predictive Analytics, Customer Support
5	SHAP (SHapley Additive exPlanations)	Assigns values to features, explaining their impact	Equitable distribution of feature importance	Computationally intensive for large models	Algorithmic Trading, Investment Strategies
6	Counterfactual Explanations	Provides alternative scenarios for understanding decisions	Intuitive for end-users	Generating meaningful counterfactuals can be challenging	Personalized Financial Advice, Credit Risk Assessment
7	Anchors	Identifies minimal input features necessary for a given prediction	Easy to understand	Limited applicability to specific model types and data	Insurance Underwriting, Mortgage Approval
8	Attention Mechanisms	Highlights relevant parts of input data	Captures important features	Interpretability challenges for non-experts	Automated Trading, Sentiment Analysis
9	Local Interpretable Model (LIM)	Trains a local interpretable model around the prediction	Provides insights for specific instances	Generalization may be limited	Personalized Financial Planning, Credit Scoring
10	Explanations as Model	Trains a separate interpretable model alongside the main model	Offers a more interpretable model	Increased computational complexity	Algorithmic Trading, Fraud Detection

DALEX (Model Agnostic Exploration, Explanation, and Learning):

DALEX is a comprehensive framework for model-agnostic XAI, facilitating the exploration of model behavior, comparison of different models, and generation of easy-to-understand visualizations [79-85]. In the financial sector, DALEX can analyze and compare the performance of various models, assisting decision-makers in selecting the most interpretable and reliable model for specific use cases [86-90].

Anchors:

Electronic copy available at: https://ssrn.com/abstract=4640316

Anchors aim to identify simple, human-understandable rules that are both sufficient and necessary for a model's prediction. Such rules can act as anchors that hold true for a particular instance, providing a clear explanation for the model's decision [91-95]. In finance, anchors can identify specific conditions or criteria significantly influencing investment recommendations or risk assessments [96-103].

Counterfactual Explanations:

Counterfactual explanations generate instances that, while similar to the original input, result in a different model prediction [104-107]. By showcasing what changes in the input would lead to an alternative outcome, counterfactual explanations help users understand the decision boundaries of the model [108-112]. In finance, this approach can be valuable for understanding how slight modifications to certain variables could impact investment decisions or risk assessments.

Challenges of implementing XAI in the financial sector

While strides have been made in developing explainable AI techniques for financial decision-making, persistent challenges underscore the need for ongoing innovation in this field.

Complexity of Financial Models:

Financial models, often intricate and involving numerous variables, pose challenges in interpreting decisions made by advanced models like deep neural networks [113-119]. Striking a balance between model complexity and interpretability remains an ongoing concern.

Trade-off between Accuracy and Explainability:

The perennial trade-off between model accuracy and explainability is evident [120-124]. Simpler models are more interpretable but may not capture systems' complexity as effectively [125-129]. Achieving the right balance is crucial to meet accuracy and explainability requirements.

Dynamic Nature of Financial Markets:

Financial markets' dynamic nature poses a challenge as traditional explanations may become outdated in a rapidly evolving landscape [130-136]. Explainable AI models must adapt in real-time to changing financial conditions.

Regulatory Compliance:

The lack of standardized guidelines on acceptable levels of explainability poses a challenge for financial institutions navigating diverse regulatory landscapes while implementing explainable AI solutions [137-143].

Future developments in XAI in finance

As AI technologies become increasingly prevalent in finance, the demand for interpretable and explainable AI systems is on the rise. This section delves into the prospective advancements in XAI within the finance sector.

Interpretable Machine Learning Models:

In the future, financial entities are expected to prioritize the development and integration of machine learning models inherently designed for interpretability. While complex models, such as deep neural networks, showcase impressive performance, their lack of interpretability poses a significant hurdle. Innovations in interpretable models, including decision trees, rule-based models, and linear models, will be instrumental in overcoming this challenge. These models not only deliver accurate predictions but also furnish transparent explanations of the factors influencing their decisions.

Hybrid Models and Ensemble Techniques:

A promising direction for XAI in finance involves incorporating interpretable components into intricate AI models. Hybrid models, amalgamating the strengths of interpretable and non-interpretable models, can strike a

Electronic copy available at: https://ssrn.com/abstract=4640316

balance between accuracy and explainability [144-149]. Ensemble techniques, which amalgamate predictions from diverse models, can enhance transparency. By combining outputs from interpretable models with those from more complex models, financial institutions can maintain accuracy while providing comprehensible explanations for decisions.

Regulatory Compliance and Ethical Considerations:

With the increasing use of AI in finance, regulatory bodies are anticipated to impose more stringent requirements for transparency and interpretability. Future developments in XAI will be motivated by the necessity to adhere to regulations like GDPR and address ethical concerns surrounding AI decision-making. Financial institutions will invest in technologies and practices ensuring that their AI systems are not only accurate but also accountable and fair, aligning with regulatory frameworks and ethical standards.

Explainability as a Service (XaaS):

A growing trend in the future of XAI is the emergence of Explainability as a Service (XaaS). This model involves third-party providers offering specialized tools and services dedicated to enhancing the interpretability of AI models. By outsourcing explainability solutions, financial institutions can concentrate on their core competencies while benefiting from state-of-the-art XAI tools. This approach can streamline the integration of interpretable models into existing systems and expedite the deployment of transparent AI in finance.

Natural Language Processing (NLP) for Explanation Generation:

Advancements in Natural Language Processing (NLP) will be pivotal in the future of XAI. Financial institutions will leverage NLP techniques to generate human-readable explanations for AI model decisions. This not only improves transparency but also facilitates communication between AI systems and end-users, making it simpler for stakeholders to understand and trust the decisions made by AI algorithms.

User-Centric Explainability:

Future developments in XAI will prioritize user-centric approaches, tailoring explanations to the specific needs and backgrounds of end-users. Financial institutions will invest in designing interfaces that present interpretable information clearly and accessibly. This may involve interactive visualizations, personalized dashboards, and user-friendly explanations to ensure that individuals, including non-experts, can comprehend and trust the decisions made by AI models.

Continuous Monitoring and Model Maintenance:

To maintain ongoing trust in AI systems, financial institutions will implement continuous monitoring and model maintenance practices. This involves tracking the performance of AI models over time, identifying shifts in data distributions, and updating models accordingly. Transparent reporting on model performance and potential biases will be essential for maintaining trust, and financial institutions will invest in automated tools for real-time monitoring and adjustment of AI systems.

Education and Training Initiatives:

The future of XAI in finance will see a focus on education and training initiatives. Financial professionals and decision-makers will need to develop a deeper understanding of AI and XAI concepts to interpret and leverage AI-driven insights effectively. Training programs and resources will be developed to empower individuals to make informed decisions based on AI recommendations, fostering a culture of trust and collaboration between humans and AI systems.

Conclusions

The financial decision-making has undergone a profound transformation with the emergence of Artificial Intelligence (AI). In recent times, the integration of AI, particularly Explainable Artificial Intelligence (XAI), has played a pivotal role in augmenting transparency and accountability within financial decision-making processes.

Electronic copy available at: https://ssrn.com/abstract=4640316

Our investigation traced the evolutionary trajectory of XAI in the financial sector, from its origins in traditional machine learning models to the more advanced deep learning algorithms. As financial institutions increasingly employ complex AI models to inform decision-making, the demand for interpretability and explainability has become paramount. XAI addresses this need by offering insights into the decision-making process of AI models, ensuring that financial professionals and stakeholders can comprehend and trust the outcomes. A cornerstone of XAI lies in the development of interpretable models that empower stakeholders to grasp the rationale behind AI-driven financial decisions. Our research underscored the significance of utilizing models such as decision trees, rule-based systems, and linear regression, which not only yield accurate predictions but also provide clear explanations for their conclusions. By demystifying the opaque nature of complex AI algorithms, interpretable models pave the way for increased transparency, enabling financial experts to validate and refine decision-making processes.

Understanding the influence of input features on AI model predictions is vital for instilling trust in financial decision-making systems. Feature importance and attribution techniques emerged as powerful tools within the XAI toolkit. Our research showcased how techniques like SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations) empower stakeholders to identify the most influential factors driving AI predictions. By quantifying the contribution of each feature, financial professionals can assess model behavior and ensure alignment with industry regulations and ethical standards. In the quest for transparency and accountability, our research emphasized the importance of fostering collaboration between humans and AI systems. XAI facilitates a symbiotic relationship, where human experts leverage the strengths of AI models while maintaining oversight and control. Addressing concerns related to bias, fairness, and accountability is imperative to prevent unintended consequences. Embracing XAI in financial decision-making is not just a technological evolution; it is a paradigm shift towards a more responsible and trustworthy financial future.

References

[1] Musleh Al-Sartawi, A. M. A., Hussainey, K., & Razzaque, A. (2022). The role of artificial intelligence in sustainable finance. In Journal of Sustainable Finance and Investment. https://doi.org/10.1080/20430795.2022.2057405

[2] Ma, M. (2022). Research on the Development of Hospital Intelligent Finance Based on Artificial Intelligence. Computational Intelligence and Neuroscience, 2022. https://doi.org/10.1155/2022/6549766

[3] Berdiyeva, O., Islam, M. U., & Saeedi, M. (2021). Artificial Intelligence in Accounting and Finance: Meta-Analysis. NUST Business Review, 3(1). https://doi.org/10.37435/nbr21032502

[4] Lee, J. (2020). Access to Finance for Artificial Intelligence Regulation in the Financial Services Industry. European Business Organization Law Review, 21(4). https://doi.org/10.1007/s40804-020-00200-0

[5] Weber, P., Carl, K. V., & Hinz, O. (2023). Applications of Explainable Artificial Intelligence in Finance—a systematic review of Finance, Information Systems, and Computer Science literature. Management Review Quarterly. https://doi.org/10.1007/s11301-023-00320-0

[6] Chen, X. Q., Ma, C. Q., Ren, Y. S., Lei, Y. T., Huynh, N. Q. A., & Narayan, S. (2023). Explainable artificial intelligence in finance: A bibliometric review. Finance Research Letters, 56. https://doi.org/10.1016/j.frl.2023.104145

[7] Mat Rahim, S. R., Mohamad, Z. Z., Abu Bakar, J., Mohsin, F. H., & Md Isa, N. (2018). Artificial Intelligence, Smart Contract and Islamic Finance. Asian Social Science, 14(2). https://doi.org/10.5539/ass.v14n2p145

[8] Khmyz, O. v., & Alekseeva, S. (2022). Artificial Intelligence in Finance. SSRN Electronic Journal. https://doi.org/10.2139/ssrn.4099638

[9] Mhlanga, D. (2020). Industry 4.0 in finance: the impact of artificial intelligence (ai) on digital financial inclusion. International Journal of Financial Studies, 8(3). https://doi.org/10.3390/ijfs8030045

[10] Buckley, R. P., Zetzsche, D. A., Arner, D. W., & Tang, B. W. (2021). Regulating Artificial Intelligence in Finance: Putting the Human in the Loop. Sydney Law Review, 43(1).

[11] Tadapaneni, N. R. (2020). Artificial Intelligence in Finance and Investment. International Journal of Innovative Research in Science, Engineering and Technology, 9(5).

Electronic copy available at: https://ssrn.com/abstract=4640316

[12] Ebrahim, H. I. J., Alaswad, H. M. A., Fadhul, S. M. J. M., & Binsaddig, R. (2023). Artificial Intelligence in Finance. In Studies in Systems, Decision and Control (Vol. 470). https://doi.org/10.1007/978-3-031-28314-7_18

[13] Černevičienė, J., & Kabašinskas, A. (2022). Review of Multi-Criteria Decision-Making Methods in Finance Using Explainable Artificial Intelligence. In Frontiers in Artificial Intelligence (Vol. 5). https://doi.org/10.3389/frai.2022.827584

[14] Giudici, P., & Raffinetti, E. (2021). Shapley-Lorenz eXplainable Artificial Intelligence. Expert Systems with Applications, 167. https://doi.org/10.1016/j.eswa.2020.114104

[15] Ghosh, I., Alfaro-Cortés, E., Gámez, M., & García-Rubio, N. (2023). Prediction and interpretation of daily NFT and DeFi prices dynamics: Inspection through ensemble machine learning & XAI. International Review of Financial Analysis, 87. https://doi.org/10.1016/j.irfa.2023.102558

[16] Saranya, A., & Subhashini, R. (2023). A systematic review of Explainable Artificial Intelligence models and applications: Recent developments and future trends. In Decision Analytics Journal (Vol. 7). https://doi.org/10.1016/j.dajour.2023.100230

[17] Confalonieri, R., Coba, L., Wagner, B., & Besold, T. R. (2021). A historical perspective of explainable Artificial Intelligence. Wiley Interdisciplinary Reviews: Data Mining and Knowledge Discovery, 11(1). https://doi.org/10.1002/widm.1391

[18] Caffo, B. S., D'Asaro, F. A., Garcez, A., & Raffinetti, E. (2022). Editorial: Explainable artificial intelligence models and methods in finance and healthcare. In Frontiers in Artificial Intelligence (Vol. 5). https://doi.org/10.3389/frai.2022.970246

[19] Çelik, T. B., İcan, Ö., & Bulut, E. (2023). Extending machine learning prediction capabilities by explainable AI in financial time series prediction. Applied Soft Computing, 132, 109876.

[20] Islam, S. R., Eberle, W., Ghafoor, S. K., & Ahmed, M. (2021). Explainable artificial intelligence approaches: A survey. arXiv preprint arXiv:2101.09429.

[21] Hanif, A. (2021). Towards Explainable Artificial Intelligence in Banking and Financial Services. arXiv preprint arXiv:2112.08441.

[22] Misheva, B. H., Osterrieder, J., Hirsa, A., Kulkarni, O., & Lin, S. F. (2021). Explainable AI in credit risk management. arXiv preprint arXiv:2103.00949.

[23] Knapič, S., Malhi, A., Saluja, R., & Främling, K. (2021). Explainable artificial intelligence for human decision support system in the medical domain. Machine Learning and Knowledge Extraction, 3(3), 740-770.

[24] Zhang, Z., Wu, C., Qu, S., & Chen, X. (2022). An explainable artificial intelligence approach for financial distress prediction. Information Processing & Management, 59(4), 102988.

[25] Sahakyan, M., Aung, Z., & Rahwan, T. (2021). Explainable artificial intelligence for tabular data: A survey. IEEE access, 9, 135392-135422.

[26] Lin, K., & Gao, Y. (2022). Model interpretability of financial fraud detection by group SHAP. Expert Systems with Applications, 210, 118354.

[27] Fritz-Morgenthal, S., Hein, B., & Papenbrock, J. (2022). Financial risk management and explainable, trustworthy, responsible AI. Frontiers in artificial intelligence, 5, 779799.

[28] Gelsomino, L. M., Sardesai, S., Pirttilä, M., & Henke, M. (2023). Addressing the relation between transparency and supply chain finance schemes. International Journal of Production Research, 61(17). https://doi.org/10.1080/00207543.2022.2115575

[29] Bukhtiarova, A., Dukhno, Y., Kulish, G., Kurochkina, I., & Lypchanskyi, V. (2019). Ensuring transparency of key public finance authorities. Investment Management and Financial Innovations, 16(2). https://doi.org/10.21511/imfi.16(2).2019.11

[30] Shkolnyk, I., Kozmenko, O., Nowacki, R., & Mershchii, B. (2020). Dependence of the state of public finances on their transparency and the level of corruption in a country. Economics and Sociology, 13(4). https://doi.org/10.14254/2071-789X.2020/13-4/18

[31] Cormier, B. (2023). Chinese or western finance? Transparency, official credit flows, and the international political economy of development. Review of International Organizations, 18(2). https://doi.org/10.1007/s11558-022-09469-x

Electronic copy available at: https://ssrn.com/abstract=4640316

[32] Shkolnyk, I., Melnyk, T., Havrysh, Y., & Ivanchenko, A. (2019). Local finance transparency in Ukraine. Public and Municipal Finance, 8(1). https://doi.org/10.21511/pmf.08(1).2019.06

[33] Tonhäuser, V., & Stavenes, T. (2020). Why change party finance transparency? Political competition and evidence from the 'deviant' case of Norway. European Journal of Political Research, 59(3). https://doi.org/10.1111/1475-6765.12369

[34] Wood, A. K., & Grose, C. R. (2022). Campaign Finance Transparency Affects Legislators' Election Outcomes and Behavior. American Journal of Political Science, 66(2). https://doi.org/10.1111/ajps.12676

[35] Wood, A. K. (2022). Voters Use Campaign Finance Transparency and Compliance Information. Political Behavior. https://doi.org/10.1007/s11109-022-09776-4

[36] Tewari, N. (2023). Artificial Intelligence in Finance and Industry: Opportunities and Challenges. In Decision Strategies and Artificial Intelligence Navigating the Business Landscape. https://doi.org/10.59646/edbookc5/009

[37] Ashraf, D. (2023). Digital Finance and Artificial Intelligence: Islamic Finance Challenges and Prospects. https://doi.org/10.1007/978-3-031-13302-2_11

[38] Deviprasad, K., & Pattanayak, A. (2022). Artificial Intelligence in Finance: Opportunities and Challenges. International Journal of Research Publication and Reviews, 04(01).

[39] Giudici, P. (2018). Fintech Risk Management: A Research Challenge for Artificial Intelligence in Finance. Frontiers in Artificial Intelligence, 1. https://doi.org/10.3389/frai.2018.00001

[40] Venerito, V., Bilgin, E., Iannone, F., & Kiraz, S. (2023). AI am a rheumatologist: a practical primer to large language models for rheumatologists. Rheumatology. https://doi.org/10.1093/rheumatology/kead291

[41] Confalonieri, R., Prado, F. M. del, Agramunt, S., Malagarriga, D., Faggion, D., Weyde, T., & Besold, T. R. (2019). An Ontology-based Approach to Explaining Artificial Neural Networks. ArXiv, 1906.08362(January).

[42] Liu, T., & Zheng, Z. (2020). Negotiation Assistant Bot of Pricing Prediction Based on Machine Learning. International Journal of Intelligence Science, 10(02). https://doi.org/10.4236/ijis.2020.102002

[43] Guidotti, R., & D'Onofrio, M. (2021). Matrix Profile-Based Interpretable Time Series Classifier. Frontiers in Artificial Intelligence, 4. https://doi.org/10.3389/frai.2021.699448

[44] Abbas, J. G., Pereira, S. L., & Mahalakshmi, V. (2025). Artificial Intelligence in Behavioural Finance Using A Sophisticated Decision-Tree Algorithm. International Journal of Electronic Finance, 1(1). https://doi.org/10.1504/ijef.2025.10057976

[45] Hessen, S. H., Abdul-Kader, H. M., Khedr, A. E., & Salem, R. K. (2022). Developing Multiagent E-Learning System-Based Machine Learning and Feature Selection Techniques. Computational Intelligence and Neuroscience, 2022. https://doi.org/10.1155/2022/2941840

[46] Ampomah, E. K., Nyame, G., Qin, Z., Addo, P. C., Gyamfi, E. O., & Gyan, M. (2021). Stock market prediction with gaussian naïve bayes machine learning algorithm. Informatica (Slovenia), 45(2). https://doi.org/10.31449/inf.v45i2.3407

[47] Li, Z. X., Geng, X. Y., Wang, J., & Zhuge, F. (2021). Emerging Artificial Neuron Devices for Probabilistic Computing. In Frontiers in Neuroscience (Vol. 15). https://doi.org/10.3389/fnins.2021.717947

[48] Peng, C. L., Chen, W. K., & Wei, A. P. (2021). Teaching capm for a pre-finance graduate program at the stem undergraduate level: Linear algebra perspective. Mathematics, 9(14). https://doi.org/10.3390/math9141668

[49] Berner, J., Grohs, P., & Jentzen, A. (2020). Analysis of the Generalization Error: Empirical Risk Minimization over Deep Artificial Neural Networks Overcomes the Curse of Dimensionality in the Numerical Approximation of Black--Scholes Partial Differential Equations. SIAM Journal on Mathematics of Data Science, 2(3). https://doi.org/10.1137/19m125649x

[50] Rane, Nitin (2023) Transformers in Intelligent Architecture, Engineering, and Construction (AEC) Industry: Applications, Challenges, and Future Scope. Available at SSRN: https://ssrn.com/abstract=4609914 or http://dx.doi.org/10.2139/ssrn.4609914

[51] Rane, Nitin (2023) Transformers in Industry 4.0, Industry 5.0, and Society 5.0: Roles and Challenges. Available at SSRN: https://ssrn.com/abstract=4609915 or http://dx.doi.org/10.2139/ssrn.4609915

Electronic copy available at: https://ssrn.com/abstract=4640316

[52] Gautam, V. K., Pande, C. B., Moharir, K. N., Varade, A. M., Rane, N. L., Egbueri, J. C., & Alshehri, F. (2023). Prediction of Sodium Hazard of Irrigation Purpose using Artificial Neural Network Modelling. Sustainability, 15(9), 7593. https://doi.org/10.3390/su15097593

[53] Rane, Nitin (2023) Integrating Leading-Edge Artificial Intelligence (AI), Internet of Things (IoT), and Big Data Technologies for Smart and Sustainable Architecture, Engineering and Construction (AEC) Industry: Challenges and Future Directions. Available at SSRN: https://ssrn.com/abstract=4616049 or http://dx.doi.org/10.2139/ssrn.4616049

[54] Rane, Nitin (2023) Transformers for Medical Image Analysis: Applications, Challenges, and Future Scope. Available at SSRN: https://ssrn.com/abstract=4622241 or http://dx.doi.org/10.2139/ssrn.4622241

[55] Luu, N. T., & Hung, P. D. (2021). Loan Default Prediction Using Artificial Intelligence for the Borrow – Lend Collaboration. Lecture Notes in Computer Science (Including Subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics), 12983 LNCS. https://doi.org/10.1007/978-3-030-88207-5_26

[56] Tavares, T. H. B. de C., Ferreira, B. P., & Mendes, E. M. A. M. (2022). Fuzzy time series model based on red–black trees for stock index forecasting[Formula presented]. Applied Soft Computing, 127. https://doi.org/10.1016/j.asoc.2022.109323

[57] Wyrobek, J. (2018). Predicting Bankruptcy at Polish Companies: A Comparison of Selected Machine Learning and Deep Learning Algorithms. Zeszyty Naukowe Uniwersytetu Ekonomicznego w Krakowie, 6(978). https://doi.org/10.15678/znuek.2018.0978.0603

[58] Liu, Y. (2023). Artificial Intelligence and Machine Learning based Financial Risk Network Assessment Model. Proceedings - 2023 12th IEEE International Conference on Communication Systems and Network Technologies, CSNT 2023. https://doi.org/10.1109/CSNT57126.2023.10134653

[59] Veeramanikandan, V., & Jeyakarthic, M. (2019). An ensemble model of outlier detection with random tree data classification for financial credit scoring prediction system. International Journal of Recent Technology and Engineering, 8(3). https://doi.org/10.35940/ijrte.C5850.098319

[60] Nabipour, M., Nayyeri, P., Jabani, H., Mosavi, A., Salwana, E., & Shahab, S. (2020). Deep learning for stock market prediction. Entropy, 22(8). https://doi.org/10.3390/E22080840

[61] Dubey, R. K., & Nizar Banu, P. K. (2018). Analysis of supervised and unsupervised technique for authentication dataset. International Journal of Engineering and Technology(UAE), 7(4). https://doi.org/10.14419/ijet.v7i4.9564

[62] Tsai, C. F., & Wu, J. W. (2008). Using neural network ensembles for bankruptcy prediction and credit scoring. Expert Systems with Applications, 34(4). https://doi.org/10.1016/j.eswa.2007.05.019

[63] Ghosh, P., Mahapatra, S., Jana, S., & Kr. Jha, R. (2023). A Study on Music Genre Classification using Machine Learning. International Journal of Engineering Business and Social Science, 1(04). https://doi.org/10.58451/ijebss.v1i04.55

[64] Urbinate, E. F., Felizardo, L. K., & Del-Moral-Hernandez, E. (2023). Deep learning stacking for financial time series forecasting: an analysis with synthetic and real-world time series. https://doi.org/10.5753/eniac.2022.227540

[65] Munkhdalai, L., Munkhdalai, T., Namsrai, O. E., Lee, J. Y., & Ryu, K. H. (2019). An empirical comparison of machine-learning methods on bank client credit assessments. Sustainability (Switzerland), 11(3). https://doi.org/10.3390/su11030699

[66] Sonkavde, G., Dharrao, D. S., Bongale, A. M., Deokate, S. T., Doreswamy, D., & Bhat, S. K. (2023). Forecasting Stock Market Prices Using Machine Learning and Deep Learning Models: A Systematic Review, Performance Analysis and Discussion of Implications. International Journal of Financial Studies, 11(3). https://doi.org/10.3390/ijfs11030094

[67] Maknickienė, N., Stankevičienė, J., & Maknickas, A. (2020). Comparison of forex market forecasting tools based on Evolino ensemble and technical analysis indicators. Romanian Journal of Economic Forecasting, 23(3).

[68] Han, D., & Ding, L. (2021). Financial Risk Prediction of Manufacturing Enterprises Based on SMOTE-Ensemble Learning. Proceedings - 2021 International Conference on Management Science and Software Engineering, ICMSSE 2021. https://doi.org/10.1109/ICMSSE53595.2021.00016

Electronic copy available at: https://ssrn.com/abstract=4640316

[69] Sayed-Mouchaweh, M. (2021). Prologue: Introduction to explainable artificial intelligence. In Explainable AI Within the Digital Transformation and Cyber Physical Systems: XAI Methods and Applications. https://doi.org/10.1007/978-3-030-76409-8_1

[70] Hadji Misheva, B., Hirsa, A., Osterrieder, J., Kulkarni, O., & Fung Lin, S. (2021). Explainable AI in Credit Risk Management. SSRN Electronic Journal. https://doi.org/10.2139/ssrn.3795322

[71] Islam, S. R., Eberle, W., & Ghafoor, S. K. (2020). Towards quantification of explainability in explainable artificial intelligence methods. Proceedings of the 33rd International Florida Artificial Intelligence Research Society Conference, FLAIRS 2020.

[72] la Gatta, V., Moscato, V., Postiglione, M., & Sperlì, G. (2021). PASTLE: Pivot-aided space transformation for local explanations. Pattern Recognition Letters, 149. https://doi.org/10.1016/j.patrec.2021.05.018

[73] Reddy, G. P., & Kumar, Y. V. P. (2023). Explainable AI (XAI): Explained. 2023 IEEE Open Conference of Electrical, Electronic and Information Sciences, EStream 2023 - Proceedings. https://doi.org/10.1109/eStream59056.2023.10134984

[74] Rane, Nitin (2023) Integrating Building Information Modelling (BIM) and Artificial Intelligence (AI) for Smart Construction Schedule, Cost, Quality, and Safety Management: Challenges and Opportunities. Available at SSRN: https://ssrn.com/abstract=4616055 or http://dx.doi.org/10.2139/ssrn.4616055

[75] Rane, Nitin (2023) ChatGPT and Similar Generative Artificial Intelligence (AI) for Building and Construction Industry: Contribution, Opportunities and Challenges of Large Language Models for Industry 4.0, Industry 5.0, and Society 5.0. Available at SSRN: https://ssrn.com/abstract=4603221 or http://dx.doi.org/10.2139/ssrn.4603221

[76] Rane, Nitin (2023) Chatbot-Enhanced Teaching and Learning: Implementation Strategies, Challenges, and the Role of ChatGPT in Education. Available at SSRN: https://ssrn.com/abstract=4603204 or http://dx.doi.org/10.2139/ssrn.4603204

[77] Rane, Nitin (2023) Enhancing Customer Loyalty through Artificial Intelligence (AI), Internet of Things (IoT), and Big Data Technologies: Improving Customer Satisfaction, Engagement, Relationship, and Experience. Available at SSRN: https://ssrn.com/abstract=4616051 or http://dx.doi.org/10.2139/ssrn.4616051

[78] Rane, Nitin (2023) 3D, 4D, and 5D printing in Architecture, Engineering, and Construction (AEC) Industry: Applications, Challenges, and Future Scope. Available at SSRN: https://ssrn.com/abstract=4609912 or http://dx.doi.org/10.2139/ssrn.4609912

[79] Gunasekaran, A., Chen, M., Hill, R., & McCabe, K. (2023). Method Agnostic Model Class Reliance (MAMCR) Explanation of Multiple Machine Learning Models. Communications in Computer and Information Science, 1788 CCIS. https://doi.org/10.1007/978-3-031-27609-5_5

[80] Mohanty, P. K., Das, P., & Roy, D. S. (2022). Predicting daily household energy usages by using Model Agnostic Language for Exploration and Explanation. Proceedings - 2022 OITS International Conference on Information Technology, OCIT 2022. https://doi.org/10.1109/OCIT56763.2022.00106

[81] Pranav, R., & Gururaja, H. S. (2023). Explainable Stacking Machine Learning Ensemble for Predicting Airline Customer Satisfaction. Lecture Notes in Networks and Systems, 608. https://doi.org/10.1007/978-981-19-9225-4_4

[82] Shaw, P., Pachpor, K., & Sankaranarayanan, S. (2022). Explainable AI Enabled Infant Mortality Prediction Based on Neonatal Sepsis. Computer Systems Science and Engineering, 44(1). https://doi.org/10.32604/csse.2023.025281

[83] Ribeiro, J., Silva, R., Cardoso, L., & Alves, R. (2021). Does Dataset Complexity Matters for Model Explainers? Proceedings - 2021 IEEE International Conference on Big Data, Big Data 2021. https://doi.org/10.1109/BigData52589.2021.9671630

[84] Cuvelier, C., Fassier, T., Dalex, M., Lagrue, S., Trombert, V., Herrmann, F. R., Zekry, D., & Roux, X. (2021). Oldest-old Covid-19 patients: Lessons learnt in a geriatric intermediate care unit. In Journal of Critical Care (Vol. 63). https://doi.org/10.1016/j.jcrc.2021.02.003

[85] Wang, Z., Zhang, L., Huang, T., Yang, R., Cheng, H., Wang, H., Yin, H., & Lyu, J. (2023). Developing an explainable machine learning model to predict the mechanical ventilation duration of patients with ARDS in intensive care units. Heart and Lung, 58. https://doi.org/10.1016/j.hrtlng.2022.11.005

Electronic copy available at: https://ssrn.com/abstract=4640316

[86] Kramar, T., Vondrous, P., Kolarikova, M., Kovanda, K., Kolarik, L., & Ondruska, M. (2015). Resistance spot welding of magnesium alloy AZ61. MM Science Journal, MAR 2015. https://doi.org/10.17973/mmsj.2015_03_201512

[87] Tatsumi, K., Igarashi, N., & Mengxue, X. (2021). Prediction of plant-level tomato biomass and yield using machine learning with unmanned aerial vehicle imagery. Plant Methods, 17(1). https://doi.org/10.1186/s13007-021-00761-2

[88] Baniecki, H., Kretowicz, W., Piatyszek, P., Wisniewski, J., & Biecek, P. (2021). dalex: Responsible machine learning with interactive explainability and fairness in python. Journal of Machine Learning Research, 22.

[89] Biecek, P. (2018). Dalex: Explainers for complex predictive models in R. Journal of Machine Learning Research, 19.

[90] Bschorr, T., Cramer, H., & Zech, F. (2011). Optimisation of the projection geometry for the resistance welding on newly developed advanced high-strength to ultrahigh-strength steel materials. Welding and Cutting, 10(2).

[91] Rane, Nitin (2023) Enhancing Mathematical Capabilities through ChatGPT and Similar Generative Artificial Intelligence: Roles and Challenges in Solving Mathematical Problems. Available at SSRN: https://ssrn.com/abstract=4603237 or http://dx.doi.org/10.2139/ssrn.4603237

[92] Rane, Nitin (2023) Transforming Structural Engineering through ChatGPT and Similar Generative Artificial Intelligence: Roles, Challenges, and Opportunities. Available at SSRN: https://ssrn.com/abstract=4603242 or http://dx.doi.org/10.2139/ssrn.4603242

[93] Rane, Nitin (2023) Roles and Challenges of ChatGPT and Similar Generative Artificial Intelligence for Achieving the Sustainable Development Goals (SDGs). Available at SSRN: https://ssrn.com/abstract=4603244 or http://dx.doi.org/10.2139/ssrn.4603244

[94] Rane, N. L. (2023). Multidisciplinary collaboration: key players in successful implementation of ChatGPT and similar generative artificial intelligence in manufacturing, finance, retail, transportation, and construction industry. https://doi.org/10.31219/osf.io/npm3d

[95] Rane, Nitin (2023) Potential Role and Challenges of ChatGPT and Similar Generative Artificial Intelligence in Architectural Engineering. Available at SSRN: https://ssrn.com/abstract=4607767 or http://dx.doi.org/10.2139/ssrn.4607767

[96] Sironi, P. (2023). Radical rationality allows coping with fundamental uncertainty on financial markets. Artificial Intelligence and Financial Behaviour: 0, 89.

[97] Archer, S., & Zumbansen, P. C. (2023). Sustainable Transformation of Business and Finance: A Democratic Challenge in An Age of Climate Change and Artificial Intelligence. McGill University Research Paper, 1.

[98] Tshimula, J. M., Nkashama, D. J. K., Owusu, P., Frappier, M., Tardif, P. M., Kabanza, F., ... & Chikhaoui, B. (2023). Characterizing Financial Market Coverage using Artificial Intelligence. arXiv preprint arXiv:2302.03694.

[99] Notovich, A., Chalutz-Ben Gal, H., & Ben-Gal, I. (2023). Explainable Artificial Intelligence (XAI): Motivation, Terminology, and Taxonomy. In Machine Learning for Data Science Handbook: Data Mining and Knowledge Discovery Handbook (pp. 971-985). Cham: Springer International Publishing.

[100] Pashentsev, E. (2023). The Malicious Use of Artificial Intelligence Through Agenda Setting. In The Palgrave Handbook of Malicious Use of AI and Psychological Security (pp. 133-172). Cham: Springer International Publishing.

[101] Band, S. S., Yarahmadi, A., Hsu, C. C., Biyari, M., Sookhak, M., Ameri, R., ... & Liang, H. W. (2023). Application of explainable artificial intelligence in medical health: A systematic review of interpretability methods. Informatics in Medicine Unlocked, 101286.

[102] wael AL-khatib, A. (2023). Drivers of generative artificial intelligence to fostering exploitative and exploratory innovation: A TOE framework. Technology in Society, 75, 102403.

[103] Li, J., Lian, Z., Wu, Z., Zeng, L., Mu, L., Yuan, Y., ... & Ye, J. (2023). Artificial intelligence–based method for the rapid detection of fish parasites (Ichthyophthirius multifiliis, Gyrodactylus kobayashii, and Argulus japonicus). Aquaculture, 563, 738790.

[104] Song, Z., Coupé, T., & Reed, W. R. (2021). Estimating the effect of the one-child policy on Chinese household savings - Evidence from an Oaxaca decomposition. China Economic Review, 66. https://doi.org/10.1016/j.chieco.2020.101570

Electronic copy available at: https://ssrn.com/abstract=4640316

[105] Prado-Romero, M. A., & Stilo, G. (2022). GRETEL: Graph Counterfactual Explanation Evaluation Framework. International Conference on Information and Knowledge Management, Proceedings. https://doi.org/10.1145/3511808.3557608

[106] Carreira-Perpiñán, M., & Hada, S. S. (2021). Counterfactual Explanations for Oblique Decision Trees: Exact, Efficient Algorithms. 35th AAAI Conference on Artificial Intelligence, AAAI 2021, 8A. https://doi.org/10.1609/aaai.v35i8.16851

[107] Wang, D., Chen, Z., Florescu, I., & Wen, B. (2023). A sparsity algorithm for finding optimal counterfactual explanations: Application to corporate credit rating. Research in International Business and Finance, 64. https://doi.org/10.1016/j.ribaf.2022.101869

[108] Rane, N. L., Achari, A., Choudhary, S. P., Mallick, S. K., Pande, C. B., Srivastava, A., & Moharir, K. (2023). A decision framework for potential dam site selection using GIS, MIF and TOPSIS in Ulhas river basin, India. Journal of Cleaner Production, 138890. https://doi.org/10.1016/j.jclepro.2023.138890

[109] Rane, N. L., Achari, A., Saha, A., Poddar, I., Rane, J., Pande, C. B., & Roy, R. (2023). An integrated GIS, MIF, and TOPSIS approach for appraising electric vehicle charging station suitability zones in Mumbai, India. Sustainable Cities and Society, 104717. https://doi.org/10.1016/j.scs.2023.104717

[110] Patil, D. R., Rane, N. L., (2023) Customer experience and satisfaction: importance of customer reviews and customer value on buying preference, International Research Journal of Modernization in Engineering Technology and Science, 5(3), 3437- 3447. https://www.doi.org/10.56726/IRJMETS36460

[111] Rane, N. L., Anand, A., Deepak K., (2023). Evaluating the Selection Criteria of Formwork System (FS) for RCC Building Construction. International Journal of Engineering Trends and Technology, vol. 71, no. 3, pp. 197-205. https://doi.org/10.14445/22315381/IJETT-V71I3P220

[112] Rane, N. L., Achari, A., Hashemizadeh, A., Phalak, S., Pande, C. B., Giduturi, M., Khan M. Y., Tolche A, D., Tamam, N., Abbas, M,, & Yadav, K. K. (2023). Identification of sustainable urban settlement sites using interrelationship based multi-influencing factor technique and GIS. Geocarto International, 1-27. https://doi.org/10.1080/10106049.2023.2272670

[113] Černevičienė, J., & Kabašinskas, A. (2022). Review of multi-criteria decision-making methods in finance using explainable artificial intelligence. Frontiers in artificial intelligence, 5, 827584.

[114] Došilović, F. K., Brčić, M., & Hlupić, N. (2018, May). Explainable artificial intelligence: A survey. In 2018 41st International convention on information and communication technology, electronics and microelectronics (MIPRO) (pp. 0210-0215). IEEE.

[115] Biecek, P., Chlebus, M., Gajda, J., Gosiewska, A., Kozak, A., Ogonowski, D., ... & Wojewnik, P. (2021). Enabling machine learning algorithms for credit scoring--explainable artificial intelligence (XAI) methods for clear understanding complex predictive models. arXiv preprint arXiv:2104.06735.

[116] Mayenberger, D. (2021). Explainable artificial intelligence: A global fast approach. Journal of Risk Management in Financial Institutions, 14(3), 287-300.

[117] Hanif, A., Zhang, X., & Wood, S. (2021, October). A survey on explainable artificial intelligence techniques and challenges. In 2021 IEEE 25th international enterprise distributed object computing workshop (EDOCW) (pp. 81-89). IEEE.

[118] Minh, D., Wang, H. X., Li, Y. F., & Nguyen, T. N. (2022). Explainable artificial intelligence: a comprehensive review. Artificial Intelligence Review, 1-66.

[119] Fritz-Morgenthal, S., Hein, B., & Papenbrock, J. (2022). Financial risk management and explainable, trustworthy, responsible AI. Frontiers in artificial intelligence, 5, 779799.

[120] Balagopalan, A., Zhang, H., Hamidieh, K., Hartvigsen, T., Rudzicz, F., & Ghassemi, M. (2022). The Road to Explainability is Paved with Bias: Measuring the Fairness of Explanations. ACM International Conference Proceeding Series. https://doi.org/10.1145/3531146.3533179

[121] Jeong, D., Park, C., Kang, T., Shin, K., Choi, S., & Hwang, J. (2023). Measuring ethics level of technological topics using phylogenetic tree. Technology Analysis and Strategic Management. https://doi.org/10.1080/09537325.2023.2209214

[122] Sovrano, F., & Vitali, F. (2023). An objective metric for Explainable AI: How and why to estimate the degree of explainability. Knowledge-Based Systems, 278. https://doi.org/10.1016/j.knosys.2023.110866

[123] Lee, K., Lee, H., Lee, H., Yoon, Y., Lee, E., & Rhee, W. (2018). Assuring explainability on demand response targeting via credit scoring. Energy, 161. https://doi.org/10.1016/j.energy.2018.07.179

Electronic copy available at: https://ssrn.com/abstract=4640316

[124] Aldoseri, A., Al-Khalifa, K. N., & Hamouda, A. M. (2023). Re-Thinking Data Strategy and Integration for Artificial Intelligence: Concepts, Opportunities, and Challenges. In Applied Sciences (Switzerland) (Vol. 13, Issue 12). https://doi.org/10.3390/app13127082

[125] Rane, N. L., & Jayaraj, G. K. (2022). Comparison of multi-influence factor, weight of evidence and frequency ratio techniques to evaluate groundwater potential zones of basaltic aquifer systems. Environment, Development and Sustainability, 24(2), 2315-2344. https://doi.org/10.1007/s10668-021-01535-5

[126] Rane, Nitin (2023) ChatGPT and Similar Generative Artificial Intelligence (AI) for Smart Industry: Role, Challenges and Opportunities for Industry 4.0, Industry 5.0 and Society 5.0. Available at SSRN: https://ssrn.com/abstract=4603234 or http://dx.doi.org/10.2139/ssrn.4603234

[127] Moharir, K. N., Pande, C. B., Gautam, V. K., Singh, S. K., & Rane, N. L. (2023). Integration of hydrogeological data, GIS and AHP techniques applied to delineate groundwater potential zones in sandstone, limestone and shales rocks of the Damoh district, (MP) central India. Environmental Research, 115832. https://doi.org/10.1016/j.envres.2023.115832

[128] Rane, Nitin (2023) Role and Challenges of ChatGPT and Similar Generative Artificial Intelligence in Business Management. Available at SSRN: https://ssrn.com/abstract=4603227 or http://dx.doi.org/10.2139/ssrn.4603227

[129] Rane, Nitin (2023) Role and Challenges of ChatGPT and Similar Generative Artificial Intelligence in Human Resource Management. Available at SSRN: https://ssrn.com/abstract=4603230 or http://dx.doi.org/10.2139/ssrn.4603230

[130] Liu, M. (2022). The driving forces of green bond market volatility and the response of the market to the COVID-19 pandemic. Economic Analysis and Policy, 75. https://doi.org/10.1016/j.eap.2022.05.012

[131] August, M., Cohen, D., Danyluk, M., Kass, A., Ponder, C. S., & Rosenman, E. (2022). Reimagining geographies of public finance. Progress in Human Geography, 46(2). https://doi.org/10.1177/03091325211054963

[132] Wu, R., & Liu, B. Y. (2023). Do climate policy uncertainty and investor sentiment drive the dynamic spillovers among green finance markets? Journal of Environmental Management, 347. https://doi.org/10.1016/j.jenvman.2023.119008

[133] Zhuang, X., & Wei, D. (2022). Asymmetric multifractality, comparative efficiency analysis of green finance markets: A dynamic study by index-based model. In Physica A: Statistical Mechanics and its Applications (Vol. 604). https://doi.org/10.1016/j.physa.2022.127949

[134] Bennett, D., Mekelburg, E., & Williams, T. H. (2023). BeFi meets DeFi: A behavioral finance approach to decentralized finance asset pricing. Research in International Business and Finance, 65. https://doi.org/10.1016/j.ribaf.2023.101939

[135] Zhang, Y. F., & Umair, M. (2023). Examining the interconnectedness of green finance: an analysis of dynamic spillover effects among green bonds, renewable energy, and carbon markets. Environmental Science and Pollution Research, 30(31). https://doi.org/10.1007/s11356-023-27870-w

[136] Block, J. H., Groh, A., Hornuf, L., Vanacker, T., & Vismara, S. (2021). The entrepreneurial finance markets of the future: a comparison of crowdfunding and initial coin offerings. Small Business Economics, 57(2). https://doi.org/10.1007/s11187-020-00330-2

[137] Mirestean, A., Farias, A., Deodoro, J., Boukherouaa, E. B., AlAjmi, K., Iskender, E., Ravikumar, R., & Shabsigh, G. (2021). Powering the Digital Economy: Opportunities and Risks of Artificial Intelligence in Finance. Departmental Papers, 2021(024). https://doi.org/10.5089/9781589063952.087

[138] Guo, H., & Polak, P. (2021). Artificial Intelligence and Financial Technology FinTech: How AI Is Being Used Under the Pandemic in 2020. In Studies in Computational Intelligence (Vol. 935). https://doi.org/10.1007/978-3-030-62796-6_9

[139] Remolina, N., & Gurrea-Martinez, A. (2023). Artificial intelligence in finance: Challenges, opportunities and regulatory developments. In Artificial Intelligence in Finance: Challenges, Opportunities and Regulatory Developments. https://doi.org/10.4337/9781803926179

[140] Cai, C., Marrone, M., & Linnenluecke, M. (2022). Trends in FinTech Research and Practice: Examining the Intersection with the Information Systems Field. Communications of the Association for Information Systems, 50(1). https://doi.org/10.17705/1CAIS.05036

Electronic copy available at: https://ssrn.com/abstract=4640316

[141] Wang, N., & Wang, K. (2022). Internet Financial Risk Management in the Context of Big Data and Artificial Intelligence. Mathematical Problems in Engineering, 2022. https://doi.org/10.1155/2022/6219489
[142] A. Basha, S., Elgammal, M. M., & Abuzayed, B. M. (2021). Online peer-to-peer lending: A review of the literature. Electronic Commerce Research and Applications, 48. https://doi.org/10.1016/j.elerap.2021.101069
[143] Bertoni, F., Bonini, S., Capizzi, V., Colombo, M. G., & Manigart, S. (2022). Digitization in the Market for Entrepreneurial Finance: Innovative Business Models and New Financing Channels. In Entrepreneurship: Theory and Practice (Vol. 46, Issue 5). https://doi.org/10.1177/10422587211038480
[144] Shahid, M., Khaliq, A., & Mohtesham, M. M. J. (2021). An Innovative Fintech Model for The Enhancement of Akhuwat Microfinance Practices. Al-Iqtishad: Jurnal Ilmu Ekonomi Syariah, 13(2). https://doi.org/10.15408/aiq.v13i2.22498
[145] Kao, Y. C., Shen, K. Y., Lee, S. T., & Shieh, J. C. P. (2022). Selecting the Fintech Strategy for Supply Chain Finance: A Hybrid Decision Approach for Banks. Mathematics, 10(14). https://doi.org/10.3390/math10142393
[146] Manjunath, C., Marimuthu, B., & Ghosh, B. (2023). Analysis of Nifty 50 index stock market trends using hybrid machine learning model in quantum finance. International Journal of Electrical and Computer Engineering, 13(3). https://doi.org/10.11591/ijece.v13i3.pp3549-3560
[147] Li, Y., Stasinakis, C., & Yeo, W. M. (2022). A Hybrid XGBoost-MLP Model for Credit Risk Assessment on Digital Supply Chain Finance. Forecasting, 4(1). https://doi.org/10.3390/forecast4010011
[148] Aboulamer, A. (2018). Adopting a circular business model improves market equity value. Thunderbird International Business Review, 60(5). https://doi.org/10.1002/tie.21922
[149] Liu, J., Yan, L., & Wang, D. (2022). A Hybrid Blockchain Model for Trusted Data of Supply Chain Finance. Wireless Personal Communications, 127(2). https://doi.org/10.1007/s11277-021-08451-x

Declarations

Funding: No funding was received.

Conflicts of interest/Competing interests: No conflict of interest.

Availability of data and material: Not applicable.

Code availability: Not applicable.

Acknowledgements: Not Applicable.

Electronic copy available at: https://ssrn.com/abstract=4640316